

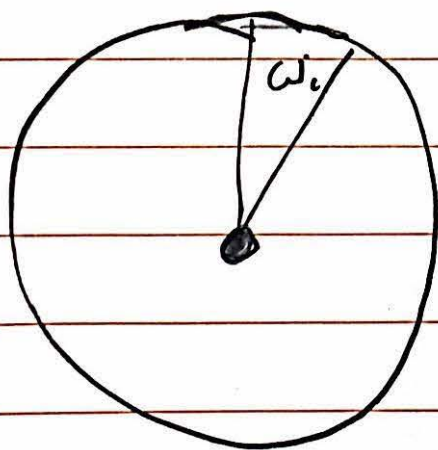
Lesson 8

Moments of Discrete Random Variables

Expectation:

We start our discussion on
expectation with an example.

Example (wheel of fortune)



Assume that the
wheel has k
slices. Slice i
wins w_i dollars.

Also, assume that we spin the wheel ~~n~~ times, and slice i appears n_i times.

Obviously

$$\sum_{i=1}^K n_i = n$$

The total amount of money that we receive is:

$$\sum_{i=1}^K n_i w_i$$

The average/per spin amount of money that we receive

is

$$\boxed{\sum_{i=1}^K \frac{n_i \omega_i}{n} = \sum_{i=1}^K \frac{n_i}{n} \omega_i} \quad \star$$

$n \rightarrow \infty \quad \frac{n_i}{n} \rightarrow p_i$ prob of slice i appearing

which, when $n \rightarrow \infty$, becomes

$$\star = \sum_{i=1}^K p_i \omega_i \quad \text{the average dollar per spin}$$

This example is a motivation

for defining the expectation

of a discrete random variable

with pmf P_X , $\# [X]$:

Def:

$$\mathbb{E}[X] = \sum_x x P_X(x)$$

Example: Bernoulli R.V.

$$P_X(0) = 1-p$$

$$P_X(1) = p$$

$$E[X] = \sum_{x=0}^1 x P_X(x) = 0 \times (1-p) + 1 \times p = p$$

Example: Zipf distribution

Assume

$$P_X(x) = \frac{C}{x^{3/2}}, \quad x \in \mathbb{N}$$

$$\sum P(x) = 1$$

$\Rightarrow$

$$C = 1 / \left(\sum_{x=1}^{\infty} \frac{1}{x^{3/2}} \right)$$

$$C =$$

$$E[X] = \sum_{x=1}^{\infty} \frac{Cx}{x^{3/2}} = \sum_{x=1}^{\infty} \frac{C}{\sqrt{x}} = +\infty$$

$$\sum_{x=1}^{\infty} \frac{1}{x^a} \begin{cases} \rightarrow \infty & a \leq 1 \\ \rightarrow < \infty & a > 1 \end{cases}$$

Therefore, the expectation of
 $X \sim \text{Lipf}(3/2)$ is infinity.

Example: $p_X(x) = \frac{C}{x^2+1}, x \in \mathbb{Z}$
 $\sum_x p_X(x) = 1$

$$C = \frac{1}{\sum_{x \in \mathbb{Z}} \frac{1}{x^2+1}}$$

$$E[X] = \sum_{x=-\infty}^{+\infty} x p_X(x)$$

$$= \sum_{x=-\infty}^{+\infty} \frac{Cx}{x^2+1}$$

= undefined



doesn't
exist

$$\lim_{T \rightarrow \infty} \left(\int_{-T}^T x dx \right) = 0$$

$$\int_{-\infty}^{+\infty} x dx = \frac{1}{2} x^2 \Big|_{-\infty}^{+\infty}$$

undefined

$$\lim_{T_1, T_2 \rightarrow \infty} \int_{-T_1}^{T_2} x dx$$

$$S_N = \sum_{n=-N}^N a_n$$

$$a_n = n$$

$$S_N = 0$$

$$\lim_{N \rightarrow \infty} S_N = 0$$

$$\lim_{N \rightarrow \infty} \sum_{n=-N}^N S_N = 0$$

$$S_{N_1, N_2} = \sum_{n=-N_1}^{+N_2} a_n \quad a_n = n$$

$$\sum_{-\infty}^{+\infty} a_n = \lim_{N_1, N_2 \rightarrow \infty} S_{N_1, N_2} \quad \text{doesn't exist}$$

This depends on how $N_1, N_2 \rightarrow \infty$

$$\left\{ \begin{array}{l} N_1 = 2N_2 \\ N_1 = N_2 \\ N_2 = 2N_1 \end{array} \right. \quad \text{Different paths, different limits}$$

$$\sum_{x=-\infty}^{+\infty} \frac{Cx}{x^2+1} \quad \text{doesn't exist.}$$

Expected Value of A Function
of A Random Variable, or
the Law of The Unconscious
Statistician (LOTUS)

We previously realized

that any real function of
a discrete r.v. is a discrete
random variable, i.e. for any
 $g: \mathbb{R} \rightarrow \mathbb{R}$, and X , $Y = g(X)$
is a R.V.

The question is how to calculate $E[Y] = E[g(X)]$. It seems that one has to calculate the pmf of Y to calculate $E[Y]$.

$$E[Y] = \sum y P_Y(y)$$

However, $E[g(X)]$ can be

calculated using the ~~pdf~~^{pmf} of X as:

$$E[Y] = E[g(X)] = \sum_x g(x) P_X(x)$$

More generally, if

$\underline{X} = (X_1, X_2, \dots, X_n)$ is a

vector of discrete r.v.'s with

joint pmf $P_{\underline{X}} = P_{X_1, X_2, \dots, X_n}$

and g is a multivariable function: $g: \mathbb{R}^n \rightarrow \mathbb{R}$,

$$E[g(\underline{X})] = \sum_{\underline{x}} g(\underline{x}) P_{X_1, \dots, X_n}(x_1, \dots, x_n)$$

where $\underline{x} = (x_1, x_2, \dots, x_n)$

$$E[\underbrace{X^2 Y^3}_{\underline{Z}}] = \sum_x \sum_y x^2 y^3 P_{X,Y}(x,y)$$

you don't need $P_{\underline{Z}}(\underline{z})$

to calculate $E[\underline{Z}]$.

Proof of LOTUS

We saw that if $Y = g(X)$,

$$P_Y(y) = P(Y=y) = \sum_{\{x | g(x)=y\}} P(X=x) \\ = \sum_{x | g(x)=y} P_X(x)$$

Then:

$$E[Y] = \sum_y y P_Y(y) \\ = \sum_y y \sum_{x | g(x)=y} P_X(x) \\ = \sum_{x | g(x)=y} g(x) P_X(x) = \sum_x g(x) P_X(x)$$

To understand the above
proof better, do it for $g(\omega) = \omega^2$

Properties of Expectation

Lemma: Let X, Y be discrete r.v.'s on $(\Omega, \mathcal{F}, P)$.

a) If $X \geq 0$ a.s. $\Rightarrow E[X] \geq 0$
 $P(X < 0) = 0$

Remark: $X \geq 0$ a.s. means:
 $P(\{\omega \mid X(\omega) \geq 0\}) = 1$

(b) If $X = c$ a.s. for some $c \in \mathbb{R}$

then $E[X] = c$

$P(\{\omega \mid X(\omega) = c\}) = 1$

$$(c) \quad a, b \in \mathbb{R}, \quad \mathbb{E}[aX + bY]$$

$$= a \mathbb{E}[X] + b \mathbb{E}[Y]$$

provided that expected values
are well-defined on both
sides (Linearity of Expectation)

Remark: The above equality
doesn't hold if some of the
expected values are not
well-defined.

Example: $P_X(x) = \frac{c}{x^2 + 1}$

$$Y = 2X, \quad Z = aX + bY$$

$$a = -2, \quad b = 1$$

$$E[Z] = E[-2X + Y]$$

$$= E[-2X + 2X] = E[0] = 0$$

but

$E[X], E[Y]$ are undefined.

(d) If $X^{\rightarrow}(\Omega) = \mathbb{Z}^{\geq 0}$ (i.e. X is

a r.v. that can take non-negative

integers), then

$$E[X] = \sum_{n \geq 0} \underbrace{P(X > n)}_{\substack{1 - F_X(n) \\ S_X(n)}}$$

Survival Function

(e) If X and Y are independent

$$\Rightarrow E[XY] = E[X]E[Y]$$

The reverse is NOT necessarily true.

(f) If $X_1, X_2, \dots, X_n$ are independent discrete r.v.'s

then

$$E\left[\prod_{i=1}^n X_i\right] = \prod_{i=1}^n E[X_i]$$

provided that all expectations are well-defined

Proof. (a), (b) are direct
consequences of definitions

$$\begin{aligned}
 (c) \quad E[aX + bY] &= \sum_{x,y} (ax + by) P_{X,Y}(x,y) \quad \text{LOTUS} \\
 &= \sum_x \sum_y (ax + by) P_{X,Y}(x,y) \\
 &= \sum_x a x \left(\sum_y P_{X,Y}(x,y) \right) + \sum_y b y \left(\sum_x P_{X,Y}(x,y) \right) \\
 &\quad \quad \quad P_X(x) \quad \quad \quad P_Y(y)
 \end{aligned}$$

$$= a \sum_x x P_X(x) + b \sum_y y P_Y(y)$$

$$= a E[X] + b E[Y]$$

$$(1) X, Y \text{ indep} \Rightarrow E[XY] = E[X]E[Y]$$

$$E[XY] \stackrel{\text{LOTUS}}{=} \sum_{x,y} xy \underbrace{P_{X,Y}(x,y)}$$

$$= \sum_x \sum_y (x) (y) (P_X(x) P_Y(y))$$

$$= \sum_x x P_X(x) \sum_y y P_Y(y)$$

$$= E[X] E[Y]$$

(d) The proof is left as an exercise. Note that

$$\sum_{n \geq 0} P(X > n) = \sum_{n \geq 0} \sum_{x > n} p_x(x)$$

DO IT!

The relationship between probability and expectation

Assume that X is a random variable (not necessarily discrete) and B is a

Borel subset of $\mathbb{R}$.

Assume that I_B is the indicator function of B

defined as
$$I_B: \mathbb{R} \rightarrow \{0, 1\}$$
$$I_B(x) = \begin{cases} 0 & x \notin B \\ 1 & x \in B \end{cases}$$

Then $Y = I_B(X)$ is a random variable (why?) and is discrete (why?) and:

$$\begin{aligned} E[I_B(X)] &= \sum_x \underbrace{I_B(x)}_{\text{LOTUS}} P_X(x) \\ &= \sum_{x \in B} P_X(x) = P(X \in B) = P_X(B) \end{aligned}$$

$$= P(\{\omega \mid X(\omega) \in B\})$$

Expectation of products of functions of random Variables.

Theoreme Let X, Y be
random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

X, Y are independent if and

only if for all functions
 $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$$

Proof:

If parts X, Y indep $\Rightarrow E[gh] = E[g] E[h]$

$$E[g(X) h(Y)] \stackrel{\text{Lorus}}{=} \sum_{x,y} g(x) h(y) P_{X,Y}(x,y)$$

$$\stackrel{\text{indep}}{=} \sum_x \sum_y g(x) h(y) P_X(x) P_Y(y)$$

$$= \sum_x g(x) P_X(x) \sum_y h(y) P_Y(y)$$

$$= E[g(X)] E[h(Y)]$$

only if part $E[gh] = E[g] E[h]$
 $\forall g, h$

$\Rightarrow X, Y$ are indep

Assume $B_1, B_2 \in \mathcal{B}(\mathbb{R})$ we need

to show that

$$P(X \in B_1, Y \in B_2) = P(X \in B_1) P(Y \in B_2)$$

$$\left. \begin{aligned} \text{Use } g(x) &= I_{B_1}(x) \\ h(y) &= I_{B_2}(y) \end{aligned} \right\}$$

$$E[I_{B_1}(x) I_{B_2}(y)] \quad \star$$

LOTUS

$$= \sum_{x,y} I_{B_1}(x) I_{B_2}(y) P_{X,Y}(x,y)$$

$$= \sum_{x \in B_1, y \in B_2} P_{X,Y}(x,y) = \underbrace{P(X \in B_1, Y \in B_2)}$$

On the other hand

$$\star = E[I_{B_1}(x)] E[I_{B_2}(y)]$$

$$= P(X \in B_1) P(Y \in B_2)$$

Moments of Discrete Random Variables

The second moment of X :

Def: If X is a discrete r.v.

and $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = \underline{x^2}$,

$$E[g(X)] = E[X^2] = \sum_x x^2 P_X(x)$$

is called the second moment of X .

The r^{th} moment:

Def: More generally, for all

$$\underline{r \in \mathbb{N}}, \quad E[X^r] = \boxed{\sum_x x^r P_X(x)}$$

$r \in \mathbb{Q}$: fractional moments

is called the r^{th} moment of X .

Central Moments

Def: $\tilde{X} = X - E[X]$ is called the centralized version

of X .

$$E[\tilde{X}] = 0$$

Def: $E[\tilde{X}^r] = E[(X - E[X])^r]$

is called the r^{th} central moment of X , for all $r \in \mathbb{N}$.

$r=2$: The second central moment is called the variance of X , $\text{Var}(X)$.

Also, the standard deviation of X , σ_X , is defined as

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Remark: If r is even, the r^{th} central moment is non-negative

(why?) $(X - E[X])^r \geq 0$
a.s.

Therefore, if the variance exists,

the standard deviation
is well-defined.

Lemma) let X be a
discrete r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$.

a) $E[X] < \infty \Rightarrow \text{Var}(X) =$ _____

$$E[X^2] - (E[X])^2$$

(b) $a \in \mathbb{R}, \text{Var}(aX) = a^2 \text{Var}(X)$

(c) If $X_1, X_2, \dots, X_n$ are independent

discrete r.v.'s then

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i)$$

provided that all expectations
are well-defined.

(d) $\text{Var}(X) = 0$ if and only
if $X = c$ a.s.

Proof:

$$a) \text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(X) = E[(X - E[X])^2]$$

$$= E[X^2 - 2E[X]X + (E[X])^2]$$

Linearity

$$\begin{aligned} & \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

b) $\text{Var}(aX)$

$$\begin{aligned} &= \mathbb{E}[(aX)^2] - (\mathbb{E}[aX])^2 \quad \text{linearity} \\ &= a^2 \mathbb{E}[X^2] - (a \mathbb{E}[X])^2 \\ &= a^2 \mathbb{E}[X^2] - a^2 (\mathbb{E}[X])^2 \end{aligned}$$

$$\begin{aligned} &= a^2 \left(\mathbb{E}[X^2] - (\mathbb{E}[X])^2 \right) \\ &= a^2 \text{Var}(X) \end{aligned}$$

(c) Proof for $n=2$. The proof for $n \geq 2$ is done by mathematical induction

$$\text{Var}(X_1 + X_2) = \mathbb{E}[(X_1 + X_2)^2]$$

$$= \mathbb{E}[(X_1 + X_2)^2]$$

$$= \mathbb{E}[X_1^2 + X_2^2 + 2X_1X_2]$$

$$= (\mathbb{E}[X_1] + \mathbb{E}[X_2])^2$$

lin.

$$= \mathbb{E}[X_1^2] + \mathbb{E}[X_2^2] + \underbrace{2\mathbb{E}[X_1X_2]}_{\text{indep} = 2\mathbb{E}[X_1]\mathbb{E}[X_2]}$$

$$= \left\{ (\mathbb{E}[X_1])^2 + (\mathbb{E}[X_2])^2 + 2\mathbb{E}[X_1]\mathbb{E}[X_2] \right\}$$

$$= \mathbb{E}[X_1^2] - (\mathbb{E}[X_1])^2$$

$$+ \mathbb{E}[X_2^2] - (\mathbb{E}[X_2])^2$$

$$= \text{Var}(X_1) + \text{Var}(X_2)$$

(Try to show $\text{Var}(X_1 - X_2) = \text{Var}(X_1) + \text{Var}(X_2)$)

Induction is left to the students

d) $X = c$ a.s. $\implies \text{Var}(X) = 0$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

$$= \mathbb{E}[c^2] - (\mathbb{E}[c])^2$$

$$= c^2 - c^2 = 0$$

$\text{Var}(X) = 0 \implies X = c$ a.s.

$$\mathbb{E}[(X - \mathbb{E}[X])^2] = 0$$

$$\Rightarrow (X - \mathbb{E}[X])^2 = 0 \quad \text{a.s.}$$

(Prove this maybe just for discrete. If a non-negative r.v. has zero expectation it is zero almost surely).

$$\Rightarrow X - \mathbb{E}[X] = 0 \quad \text{a.s.}$$

$$X = \underbrace{\mathbb{E}[X]}_c \quad \text{a.s.}$$

$$\text{If } E[Y^2] = 0$$

$$E[Y^2] = \sum_y y^2 P_Y(y) = 0$$

$$y^2 = 0 \text{ when } P_Y(y) \neq 0$$

$$\Rightarrow P(Y^2 = 0) = 1$$

$$\Rightarrow P(Y = 0) = 1$$

$$\Rightarrow Y = 0 \quad \text{a.s.}$$

Skewness and Kurtosis

Def: Let X be a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$\mathbb{E} \left[\frac{(X - \overset{\mu_X}{\mathbb{E}[X]})^3}{\sigma_X^3} \right]$$

is $\mathbb{E} \left[\tilde{X}^3 / \sigma_X^3 \right]$ is

called the skewness of X .

Def: Let X be a r.v. on $(\Omega, \mathcal{F}, \mathbb{P})$. Then

$$\mathbb{E} \left[\frac{(X - \mathbb{E}[X])^4}{\sigma_X^4} \right] =$$

$E \left[\frac{\tilde{X}^4}{(\text{var}(X))^2} \right]$ is called
the kurtosis of X .

Expectation and Variance
of Important Discrete
R.V.'s:

(a) Bernoulli

$$E[X] = (1-p)x_0 + px_1 = p$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = 0^2 \times (1-p) + 1^2 \times p = p$$

$$\text{Var}(X) = p - p^2 = p \underbrace{(1-p)}_q = pq$$

$$\text{Or } \text{Var}(X) = E[(X - E[X])^2]$$

$$E[X] = p$$

$$(0-p)^2 \times (1-p) + (1-p)^2 \times p$$

$$= p^2(1-p) + p(1-p)^2$$

$$= p(1-p)[p + (1-p)] = p(1-p)$$

$$(b) \quad X \sim \text{Bin}(n, p)$$

X can be seen as the
sum of n independent Bernoulli r.v.s, i.e.

$$X_i \sim \text{Ber}(p)$$

$$X = X_1 + X_2 + \dots + X_n$$

$$E[X] = E[X_1 + X_2 + \dots + X_n]$$

$$= E[X_1] + E[X_2] + \dots + E[X_n]$$

$$= p + p + \dots + p = np$$

$$\text{Also, } \text{Var}(X) = \text{Var}\left(\sum_{i=1}^n X_i\right)$$

$$= \sum_{i=1}^n \text{Var}(X_i) = \underbrace{np(1-p)}_{npq} = npq$$

(c) Geometric r.v.

$$X \sim \text{Geo}_1(p)$$

$$p_X(x) = (1-p)^{x-1} p \quad x \in \mathbb{N}$$

$$E[X] = \sum_{x=1}^{\infty} x (1-p)^{x-1} p$$

Note that $\sum_{x=0}^{\infty} (1-p)^x = \frac{(1-p)^0}{1-(1-p)} = \frac{1}{p}$

take derivatives with respect to p :

$$\frac{d}{dp} \sum_{x=0}^{\infty} (1-p)^x = \frac{d}{dp} \left(\frac{1}{p} \right)$$

$$\sum_{x=0}^{\infty} \frac{d}{dp} (1-p)^x = -\frac{1}{p^2}$$

$$\sum_{x=0}^{\infty} -x (1-p)^{x-1} = -\frac{1}{p^2}$$

$$\sum_{x=1}^{\infty} x(1-p)^{x-1} = \frac{1}{p^2}$$

$$\sum_{x=1}^{\infty} x(1-p)^{x-1} p = \frac{1}{p}$$

$$E[X] = \frac{1}{p}$$

Second Method

$$E[X] = \sum_{n=0}^{\infty} P(X > n)$$

$$S_X(n) = 1 - P(X \leq n) = 1 - F_X(n)$$

$$S_X(n) = P(X > n) = \sum_{x=n+1}^{\infty} (1-p)^{x-1} p$$

$$= p \sum_{x=n+1}^{\infty} (1-p)^{x-1} = p \frac{(1-p)^{n+1-1}}{1-(1-p)}$$

$$= (1-p)^n$$

(Note: the CDF is $1 - S_X(n)$)

$$\text{So } F_X(n) = 1 - (1-p)^n \quad \forall n \in \mathbb{N}$$

$$E[X] = \sum_{n=0}^{\infty} (1-p)^n$$

$$= \frac{(1-p)^0}{1 - (1-p)} = \frac{1}{p}$$

To calculate the variance of $X \sim \text{Geo}_1(p)$, one can calculate

$$E[X(X+1)] = \sum_{x=1}^{\infty} x(x+1)p(1-p)^{x-1}$$

$$\sum_{x=0}^{\infty} (1-p)^x = 1/p$$

$$\frac{d}{dp} \sum_{x=0}^{\infty} x(1-p)^{x-1} = \left(\frac{1}{p^2} \right)$$

$$\frac{d}{dp} \sum_{x=0}^{\infty} -x(x-1)(1-p)^{x-2} = -\frac{2}{p^3}$$

$$\Rightarrow \sum_{x=0}^{\infty} x(x-1)(1-p)^{x-2} = \frac{2}{p^3}$$

$$x-1 = t \quad \Rightarrow \quad x = \underline{t+1}$$

$$\sum_{t=1}^{\infty} (t+1)t(1-p)^{t-1} = \frac{2}{p^3}$$

$$\sum_{t=1}^{\infty} t(t+1)p(1-p)^{t-1} = \frac{2}{p^2} \checkmark$$

$$\Rightarrow E[X(X+1)] = \frac{2}{p^2}$$

$$\Rightarrow E[X^2 + X] = \frac{2}{p^2}$$

$$E[X^2] + \underbrace{E[X]}_{1/p} = \frac{2}{p^2}$$

$$\Rightarrow E[X^2] = \frac{2}{p^2} - \frac{1}{p} = \frac{2-p}{p^2}$$

$$\text{Var}(X) = E[X^2] - [E[X]]^2$$

$$= \frac{2}{p^2} - \frac{1}{p} - \frac{1}{p^2} = \frac{1}{p^2} - \frac{1}{p} = \frac{1-p}{p^2} \checkmark$$

A rectangular box with a brown border. The top-right corner is folded over, creating a triangular flap. Inside the box, there are ten horizontal lines for writing, evenly spaced.

A rectangular box with a brown border. Inside the box, there are ten horizontal lines for writing, evenly spaced.

The calculation of $E[X]$
and $E[X^2]$ for $X \sim \text{Geo}_0(p)$
are left as an exercise.

$$\text{Geo}_0 = \text{Geo}_1 - 1$$

(d) $X \sim \text{Pois}(\lambda)$

$$P_X(x) = e^{-\lambda} \frac{\lambda^x}{x!} \quad x=0, 1, \dots$$

$$E[X] = \sum_{x=0}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!}$$

$$= \sum_{x=1}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!} = \sum_{x=1}^{\infty} e^{-\lambda} \frac{\lambda^x}{(x-1)!}$$

$$x-1 = t$$

$$= \sum_{t=0}^{\infty} e^{-\lambda} \frac{\lambda^{t+1}}{t!}$$

$$= \lambda e^{-\lambda} \left(\sum_{t=0}^{\infty} \frac{\lambda^t}{t!} \right) = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

To calculate $\text{Var}(X)$, we

first calculate $E[X(X-1)]$:

$$E[X(X-1)]$$

$$= \sum_{x=2}^{\infty} x(x-1) e^{-\lambda} \frac{\lambda^x}{x!}$$

$$= \sum_{x=2}^{\infty} x(x-1) e^{-\lambda} \frac{\lambda^x}{x!}$$

$$= \sum_{x=2}^{\infty} e^{-\lambda} \frac{\lambda^x}{(x-2)!}$$

$$x-2 = t$$

$$= \sum_{t=0}^{\infty} e^{-\lambda} \frac{\lambda^{t+2}}{t!} = \lambda^2$$

$$\mathbb{E}[X(X-1)] = \lambda^2$$

$$= \mathbb{E}[X^2] - \underbrace{\mathbb{E}[X]^2}_{\lambda^2}$$

$$\mathbb{E}[X^2] = \lambda^2 + \lambda$$

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda^2 + \lambda - \lambda^2 \\ &= \lambda \end{aligned}$$

Exercise: A negative binomial

random variable $X \sim \text{NegB}(k, p)$

can be seen as ^{the sum of} $k \text{ Geo}_1(p)$ independent

r.v.'s. Use this to calculate the

expectation and variance of X .

$$\mathbb{E}[X] = \mathbb{E}[X_1 + \dots + X_k]$$

$$X_i \sim \text{Geo}_1^p$$

$$= \mathbb{E}[X_1] + \mathbb{E}[X_2] + \dots + \mathbb{E}[X_k]$$

$$= \frac{k}{p}$$

$$\text{Var}(X) = \text{Var}(X_1) + \dots + \text{Var}(X_k)$$

$$= \frac{k(1-p)}{p^2}$$

Example: Let X and Y be independent r.v.'s with $X \sim \text{Pois}(\lambda)$ and $Y \sim \text{Geo}_1(p)$.

Find $E[X^2 Y + X Y^2]$

(Functions of Indep. r.v.'s)

$$E[X^2 Y + X Y^2]$$

$$= E[X^2 Y] + E[X Y^2]$$

$$= E[X^2] E[Y] + E[X] E[Y^2]$$

$$= (\lambda^2 + \lambda) \frac{1}{p} + \lambda \left(\frac{2-p}{p^2} \right)$$

$$E[Y^2] = \underbrace{\text{Var}(Y)}_{\frac{1-p}{p^2}} + \underbrace{(E[Y])^2}_{\left(\frac{1}{p}\right)^2}$$

A rectangular box with a brown border and a folded top-right corner, containing ten horizontal lines for writing.

A rectangular box with a brown border, containing ten horizontal lines for writing.

Exercise: Stein's Characterization for Poisson Random Variables

Assume that $X \sim \text{Pois}(\lambda)$ and

$f: \mathbb{R} \rightarrow \mathbb{R}$. Then $\mathbb{E}[X \cdot f(X)] = \lambda \mathbb{E}[f(X+1)]$

provided that all expectations

exist.

$$\mathbb{E}[X f(X)] = \sum_{x=0}^{\infty} x f(x) \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=1}^{\infty} x f(x) \frac{e^{-\lambda} \lambda^x}{x(x-1)!}$$

$$= \sum_{x=1}^{\infty} f(x) \frac{e^{-\lambda} \lambda^x}{(x-1)!}$$

$$x-1 = t$$

$$= \sum_{t=0}^{\infty} \frac{f(t+1) e^{-\lambda} \lambda^{t+1}}{t!}$$

$$= \lambda \sum_{t=0}^{\infty} \frac{f(t+1) e^{-\lambda} \lambda^t}{t!}$$

$$P_X(t)$$

$$= \lambda E[f(X+1)]$$

$$(e) \quad X \sim \text{Pow}(\alpha)$$

$$p_X(x) = \frac{1}{x^\alpha} - \frac{1}{(x+1)^\alpha} \quad \alpha > 0$$

$$E[X] = \sum_{n=0}^{\infty} P(X > n) \quad \leftarrow$$

$$P(X > n) = \sum_{x=n+1}^{\infty} \left(\frac{1}{x^\alpha} - \frac{1}{(x+1)^\alpha} \right) \quad S_X(n)$$

$$= \frac{1}{(n+1)^\alpha} - \frac{1}{(n+2)^\alpha} + \frac{1}{(n+2)^\alpha} - \frac{1}{(n+3)^\alpha} + \dots$$

$$= \frac{1}{(n+1)^\alpha}$$

$$E[X] = \sum_{n=0}^{\infty} \frac{1}{(n+1)^\alpha} = \sum_{n=1}^{\infty} \frac{1}{n^\alpha}$$

$$= \zeta(\alpha) \quad \text{finite if } \alpha > 1$$

A rectangular box with a brown border and a folded top-right corner, containing ten horizontal lines for writing.

A rectangular box with a brown border, containing ten horizontal lines for writing.

A rectangular box with a brown border and a folded top-right corner, containing 12 horizontal lines for writing.

A rectangular box with a brown border, containing 12 horizontal lines for writing.

Existence of Expected Values

The following diagram shows all different possibilities for existence of expectation of a random variable

Expectation of A R.V.

Well-defined

not well-defined
(undefined)

finite

Infinite

$$P_X(x) = \frac{C}{x^2+1}$$

$$x \in \mathbb{Z}$$

$$\sum_{x \in \mathbb{Z}} x P_X(x)$$

$$\sum_{x < 0} \frac{x C}{x^2+1} = -\infty$$

$$\sum_{x > 0} \frac{C x}{x^2+1} = +\infty$$

— Lemma: For every ^{discrete} random variable X on $(\Omega, \mathcal{F}, P)$.

(a) $E[X]$ is undefined if and only if

$\sum_{x < 0} x P_X(x)$ and $\sum_{x > 0} x P_X(x) > 0$
are both infinite.
 $\infty - \infty$

(b) $E[X]$ is well-defined

and finite if and only if

both of the sums $\sum_{x < 0} x P_X(x)$

and $\sum_{x > 0} x P_X(x)$ are both finite

(c) $E[X]$ is well-defined ^{and finite} if and
only if $E[|X|] = \sum_x |x| p_X(x)$ is
 finite.

Remark: Random variables that
 have this property are

called absolutely integrable
 (summable)

(d) : $E[X]$ is well-defined
 and infinite iff exactly one of
 $\sum_{x>0} x p_X(x)$ and $\sum_{x<0} x p_X(x)$ is

infinite

(e) $E[X^2]$ is always well defined

(finite or infinite) $\sum_{x=-\infty}^{+\infty} x^2 P_X(x)$

Remark: If $E[X^2] < \infty$, we

call X square integrable (summable).

Proof: We only show (c) and

(e),

(c) Remember if $\sum_n |a_n|$ is

finite, then $\sum_n a_n$ is finite

(because $a_n \leq |a_n|$)

$$\sum_n a_n \leq \sum_n |a_n| < \infty$$

Therefore $E[|X|] < \infty \Rightarrow E[X] < \infty$

because

$$E[|X|] = \sum_x |x| P_X(x) = \sum_x |x P_X(x)|$$

when it is finite,

$\sum_x x P_X(x)$ is also finite,

which is $E[X]$.

On the other hand, if

$E[X]$ is finite, both $\sum_{x < 0} x P_X(x)$

and $\sum_{x > 0} x P_X(x)$ are finite (see part a)

$$\sum_{x < 0} x P_X(x) \text{ finite} \Rightarrow - \sum_{x < 0} x P_X(x) \text{ finite}$$

$$\Rightarrow \sum_{x < 0} -x P_X(x) \text{ finite}$$

$$\sum_{x>0} x p_x(x) < \infty \Rightarrow \sum_x |x| p_x(x) < \infty$$

$$+ \sum_{x<0} -x p_x(x) < \infty$$

(e) $\sum_{x>0} x^2 p_x(x)$ is finite or infinite
($+\infty$)

$\sum_{x<0} x^2 p_x(x)$ is finite or infinite

but cannot be $-\infty$, so

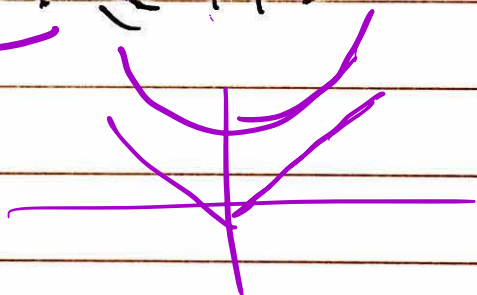
$E[X^2]$ is always well
defined.

Def: : A random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ is called square integrable iff $\mathbb{E}[X^2] < \infty$.

Lemma: A square integrable random variable is absolute integrable;

therefore, has a finite expectation.

Proof: $|x| \leq 1 + x^2$



$$\sum |x| p_X(x) \leq \sum (1 + x^2) p_X(x)$$

$$\leq \infty$$

$$= \sum p_X(x) + \sum x^2 p_X(x)$$

$$\leq \infty$$

It means

$$E[X^2] < \infty \Rightarrow E[|X|] < \infty$$

$$2) E[X] < \infty$$

The reverse is not true.

There are r.v.'s that are integrable but not square integrable.

$$P_X(x) = \frac{C}{x^3+1} \quad x \in \mathbb{N}$$

$$E[X] = \sum_{x=1}^{\infty} \frac{Cx}{x^3+1} < \infty$$

$$E[X^2] = \sum_{x=1}^{\infty} \frac{Cx^2}{x^3+1} = \infty$$

Lemma (Existence and Finiteness
of Variance) $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

a) If X is square integrable
then X has well-defined & finite
variance

(b) If X is integrable

but not square integrable
then X has well-defined and infinite
variance

(c) If X is not integrable
then the variance of X
is undefined

Proof: Obvious.

Exercise =

$$P_X(x) = \frac{C}{x^\alpha + 1} \quad \begin{array}{l} x \geq 0 \\ \alpha > 1 \end{array}$$

When does X have finite,
infinite, or undefined

variance? Repeat the
exercise with $x \in \mathbb{Z}$.

Covariance and Correlation

Def:

Assume that X, Y are

square integrable r.v.'s on $(\Omega, \mathcal{F}, \mathbb{P})$.

The Covariance of X and Y is defined as:

$$\text{Cov}(X, Y) = \mathbb{E} [$$

(centralized X, Y): $= \mathbb{E} [$

When $\text{Cov}(X, Y) = 0$, X, Y are

called uncorrelated

Lemma (Properties of Covariance)

$$(a) \text{Cor}(X, X) = \text{Var}(X)$$

$$(b) \forall a \in \mathbb{R} \text{Cor}(X, Y + a) = \text{Cor}(X, Y)$$

$$(c) \forall a, b \in \mathbb{R} \text{Cor}(X, aY + bZ) \\ = a \text{Cor}(X, Y) + b \text{Cor}(X, Z)$$

$$(d) \text{Cor}(X, Y) = \text{Cor}(Y, X)$$

Remark : Any function $f(x, y)$

that is linear with respect to

both of its ~~for~~ variables is called

a bilinear function,

$$f(x, ay + bz) =$$

$$f(ax + by, z)$$

(e) (Alternative formula for covariance)

$$\text{Cov}(X, Y) =$$

(f) (Variance of Sum of R.V.'s)
If X_1, X_2 have finite variance
 $\text{Var}(X_1 + X_2) =$

(g) If $X_1, X_2, \dots, X_n$ have finite
variance:

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n X_i\right) &= \sum_{i=1}^n \text{Var}(X_i) \\ &+ 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{Cov}(X_i, X_j) \end{aligned}$$

Example:

$$\text{Var}(X_1 + X_2 + X_3) =$$

Proof: (f, g):

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A rectangular box with a brown border, containing ten horizontal red lines for writing.

Important Note:

If X, Y are independent then they are uncorrelated, because

$$\text{Cor}(X, Y) = E[XY] - E[X]E[Y]$$

The reverse is NOT TRUE in general.

Example : Assume that

$$\begin{aligned} & P_{X,Y}(1,0) = P_{X,Y}(0,1) = P_{X,Y}(-1,0) = P_{X,Y}(0,-1) \\ & = P_{X,Y}(2,0) = P_{X,Y}(0,3) = P_{X,Y}(-2,0) = P_{X,Y}(0,-3) \end{aligned}$$

$$= 1/8$$

$$E[X] =$$

$$E[Y] =$$

$$E[XY] =$$

$$\text{Cor}(X, Y) =$$

$\Rightarrow X, Y$ are

Are X, Y independent?

$$P(X=0) = P_X(0) =$$

$$P(Y=0) = P_Y(0) =$$

$$P_{XY}(0,0) = P(X=0, Y=0) =$$

Lemma: (The Cauchy-Schwartz
- Bunyakorsky Inequality
for Expectation)

$$|E[XY]| \leq \sqrt{E[X^2]} \sqrt{E[Y^2]}$$

Proof: See Supplement.

Correlation Coefficient

Def: Assume that X and Y are two random variables with non-zero and finite variances.

The correlation coefficient

of X and Y is defined as:

$$\rho(X, Y) =$$

Theorem: Let X and Y be random variables with positive variances $\text{Var}(X) > 0$, $\text{Var}(Y) > 0$

(a) $-1 \leq \rho(X, Y) \leq 1$

(b) $\rho(X, Y) = 1$ if and only if

$\tilde{Y} = a \tilde{X}$ and $a > 0$.

(c) $\rho(X, Y) = -1$ if and only

if $\tilde{Y} = a \tilde{X}$ and $a < 0$.

A rectangular box with a brown border and a folded top-right corner. Inside the box, there are ten horizontal lines for writing, evenly spaced.

A rectangular box with a brown border. Inside the box, there are ten horizontal lines for writing, evenly spaced.

Exercise : Assume that

$X \sim \text{Bin}(n, p)$ and

$Y = n - X$. Find $P(X, Y)$.